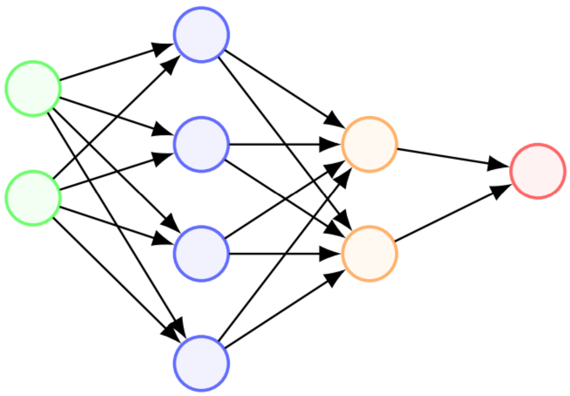
Quiz 1

Student Name: ______________________________ **Points:** 30

A Number: ____

1. Consider a Neural Network diagram:



   If we are supplying two samples with two featurs (green bubbles), then what is the dimension of Ws responsible for compressing 4 meta features to 2 meta features? For this assume that features in a feature matrix are along the column dimenions, and samples are along row dimensions.

   (a) $W_{4\times 2}$

   (b) $W_{2\times 4}$

   (c) $W_{4\times 4}$

   (d) $W_{2\times 2}$

2. Consider an activation function as $\sigma(\mathbf{x}) = \frac{1}{1+e^{-\mathbf{x}}}$, if input is $\mathbf{x} = [1.09, 2.01, 3.0]$, the what is the dimension of $A = \sigma(\mathbf{x})$?

   (a) 3×3

   (b) 1

   (c) Cannot be determined

   (d) 3

3. What does `curl` command do in a Linux machine?

   (a) It installs Python.

   (b) It searches websites for data.

   (c) It transfers data from server to client.
   (d) It creates a python virtual environment.

4. Which Linux command is used for create a new directory?

   (a) cd
   (b) mkdir
   (c) ls
   (d) mv

5. The function

$$g(z) = \begin{cases} 1, & z \geq \theta \\ 0, & otherwise \end{cases}$$

   is fully differentiable.

   (a) True
   (b) False

6. Maximum value of the Sigmoid function $\sigma(\mathbf{x}) = \frac{1}{1+e^{-x}}$ is

   (a) -1
   (b) +1
   (c) 0
   (d) 0.5

7. Consider a Neural Network definition:

```python
class QuizModel(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc1 = nn.Linear(10, 5)
        self.relu = nn.ReLU()
        self.fc2 = nn.Linear(5, 2)

    def forward(self, x):
        x = self.fc1(x)
        x = self.relu(x)
        x = self.fc2(x)
        return x
```

   What is the size of the final output?

   (a) 2
   (b) 5
   (c) 1

(d) 2×5

8. Consider

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 6 \end{bmatrix}$$

and

$$\mathbf{y} = \mathbf{f}(\mathbf{x}) = \begin{bmatrix} x_1 x_2 \\ x_1 + x_2 \\ x_1 x_2 + x_2 x_3 \end{bmatrix}$$

What is the size of Jacobian J?

(a) 4×4
(b) 4×3
(c) 3×4
(d) 3×3

9. Dropout

(a) randomly ignores a few neurons to create subnetwork on which training happens
(b) randomly ignores a few samples to train on a subset of data samples
(c) randomly ignores a few weights to train using remaining weights
(d) randomly ignores a few layers to train using remaining layers

10. A higher error on test set compared to training set denotes

(a) A lack of generalizability of the trained model
(b) Underfitting
(c) Overfitting
(d) Both (A) an (C)